Triadic Casimir — PKLSA v0.3.3 Framework Report

SST Workbench / Unregistered instance

October 18, 2026

1 Instance identity

A065-v0.3.3 versioned Triadic-Casimir research instance.

2 Research question

Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

3 Null and alternative hypotheses

No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

4 Explicit assumptions

Time-dependent regularized filament RK4 is exploratory and is not full Euler closure.

5 Definitions

u material velocity; omega=curl u; S symmetric strain.

6 Analytical equations

The official AUTO_SCIENCE_CONTRACT fragment records frozen equations.

Machine-readable science contract

Research question Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

Objective Independently verify three dataset seals, analytic Euler/gauge/sheet controls and one sourced KnotPlot trefoil finite-core geometric response without importing target force.

H_0 No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

H_1 Independently qualified PKLSA Euler motion, boundary conditions and microscopic link coupling produce held-out physical predictions.

Registered equations

E1: Incompressible Euler kinematics

$$\nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{1}$$

Dimensional check: divergence and vorticity both inverse seconds

E2: Pressure Poisson source

$$\nabla^2 p = \rho_M(\frac{1}{2}|\omega|^2 - ||S||_F^2) \tag{2}$$

Dimensional check: kg/(mˆ3 sˆ2) on each side

E3: Regularized geometric velocity diagnostic

$$\mathbf{u}(\mathbf{x}) = \frac{1}{4\pi}\sum_j \frac{\Delta\mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+1/2})}{(|\mathbf{x} - \mathbf{X}_{j+1/2}|^2 + \varepsilon^2)^{3/2}} \tag{3}$$

Dimensional check: dimensionless Gamma=1 geometry diagnostic

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify all three source seals before any scientific result is evaluated
2. **S2** [G2; equations E2, E3]: Evaluate analytic Euler controls and preregistered trefoil geometry resolution scan
3. **S3** [G3; equations E3]: Withhold physical Casimir claims without independent providers, wall dynamics, and atom-link normalization
4. **S4** [G10; equations E3]: Verify fixed-core 150/300/600 convergence, separately retaining original G2 failures
5. **S6** [G3; equations E1, E2, E3]: Evaluate G3 only after precommitted G1+G10 and temporal/analytic guard; retain G2=FAIL and physical UNRESOLVED
6. **S5** [G11; equations E3]: Verify exact Fremlin raw hash and native C++ time-series refinement; never infer real Euler solution

7 Scientific algorithm

Verify three seals; compute controls; evaluate trefoil diagnostic.

8 Dataset provenance

D01 D02 D03, authentic KnotPlot N300, and optional raw-hash qualified Fremlin native simulated time series.

9 Independent gates

Numerical refinement may unblock G3 evaluation, but never physical conclusions.

10 Numerical controls

Dimensionless finite-segment Biot-Savart and finite-difference derivatives.

11 Backend authority

CPU profile enables strict C++ gate; G3 may be evaluated but physical validation remains unresolved.

12 Blindness and integrity

All three dataset SHA256 seals checked; reveal comparator not used in blind kernel.

13 Statistics

Geometry subsamples are not independent upstream providers.

14 Result status

Read dynamic AUTO_RESULTS fragment; no target-matched force.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	c158d143c1cf453185e162d3e61fcf1d3a014ca0957b08a37e9bf56745252d2e
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	c5efb03fb8622b0d34cca8b4ca9cf4aa12535d05af5e68ac671399039cd23d02

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Are all preregistered sources present, identified and provenance-qualified?
G2	FAIL	Do reference numerics satisfy analytic/sanity and dimensional checks?
G10	PASS	Does refined KnotPlot spatial RK4 filament discretization pass 150/300 and 300/60
G11	FAIL	Does authentic Fremlin C++17 provider satisfy source, spatial and time checks (sin
G5	PASS	Does actual C++17 FP64 filament velocity match Python at 75/150/300? Kernel-o
G3	UNRESOLVED	Does target-free discovery reject the preregistered null under the registered statistic
G4	NOT_RUN_PREREQUISITE	Does an independent held-out/provenance-family confirmation reproduce the discov
G6	NOT_APPLICABLE	If enabled, does GPU screening pass CPU to GPU parity without being promoted b
G7	NOT_APPLICABLE	Does the result replicate across preregistered source families/encodings without pse
G8	NOT_RUN_PREREQUISITE	Are physical PKLSA Euler, material, surface and independent-provider closure requ
G9	DEFERRED	After all blind gates, does reveal verification match the preregistered commitment a

Backend manifest


```
{
  "native_kernel": {
    "authority": "KERNEL_ONLY_NOT_FRAMEWORK_G5",
    "backend_executed": "cpp17-serial-fp64",
    "comparisons": 1575,
    "compiler": "g++",
    "expected_component_count": 1575,
    "max_abs_component_error": 0.0,
    "max_relative_component_error": 0.0,
    "meaning": "Velocity-kernel parity only. NOT full RK4, Euler PDE, physical G5 or PKLSA dynamics.",
    "requested_backend": "cpp-17-binary64",
    "rms_component_error": 0.0,
    "schema": "A065-CPP-KERNEL-QUALIFICATION-1",
    "source_sha256": "11bec24153e8471f94e43dfeb0bae7b675fcb84a240e7d72092d08da7528fabd",
    "status": "PASS",
    "threshold": 5e-11
  },
  "physical_force_status": "UNRESOLVED",
  "pklsa_geometry": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
  "provider_manifest_sha256": "74cc9349324a15f4c1d4ca1a6910845af45aff69512538ec362ea4ae674383dc",
  "provider_status": "PASS_RAW_HASH_AND_XYZ_PARSER_ONLY",
  "python_reference": {
    "actual_backend": "python-stdlib-binary64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
```

```
  "scope": "Kernel-only native gate; complete coupled physical validation withheld",
  "sealed_source_groups": [
    "D01_EULER",
    "D02_CLOCK_LINK",
    "D03_TWO_SURFACE"
  ],
  "v033_numerical_routing": {
    "expected_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "g10_status": "PASS",
    "g3_permitted_outcome": "UNRESOLVED",
    "historical_G2": "FAIL_RETAINED",
    "observed_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "physical_force_status": "UNRESOLVED",
    "prereg_sha256": "3856a587779d82df3b11ad1e64c4e4a45b6b4318e30c00676e4015d4df841fd6",
    "protected_mismatches": [],
    "ready_for_G3_evaluation": true,
    "reference_count": 19,
    "release": "v0.3.3",
    "schema": "A065-ROUTING-DECISION-V033-1",
    "scope": "Execution-eligibility diagnostics only. NOT Casimir, gravity, full Euler or PKLSA physical
    "time_discretization": "PASS"
  }
}
```


Source manifest


```
{
  "manifest_sha256": "dc28c5220b92603e0de13b95c084b8cc23f755e25e8f0a806a13e276a98dd7d3",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A046",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-euler",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-euler",
      "registry_hint": {
        "catalog_id": "A046",
        "hits": []
      },
      "role": "analytic_control",
      "sha256": "9c4da362ab58a546fe8aff349a024dcc7fb63218ef30053dceee3d6659c574ad",
      "source_id": "D01",
      "version": "v0.3.3"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot-relaxed-3_1",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "knotplot-relaxed-3_1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
```

```
      },
      "role": "geometry_diagnostic",
      "sha256": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
      "source_id": "PKLSA",
      "version": "v0.3.3"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-link",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-link",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "toy_control",
      "sha256": "8e4bcfcdf186e539f0cdeafd9cbbf9f31dbe7bbfb95c14d89099e6c31f760af8",
      "source_id": "D02",
      "version": "v0.3.3"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-sheet",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-sheet",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "heldout_control",
      "sha256": "fff8860c2597840d0f15906a43f8710793c0909ebfdde1b1eb6f64aea7863adc",
      "source_id": "D03",
      "version": "v0.3.3"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Cas
}
```


15 Scientific interpretation

Physical Casimir force and gravitation remain unvalidated.

16 Reproduction

Run native FREEZE and FULL via the installation bridge.